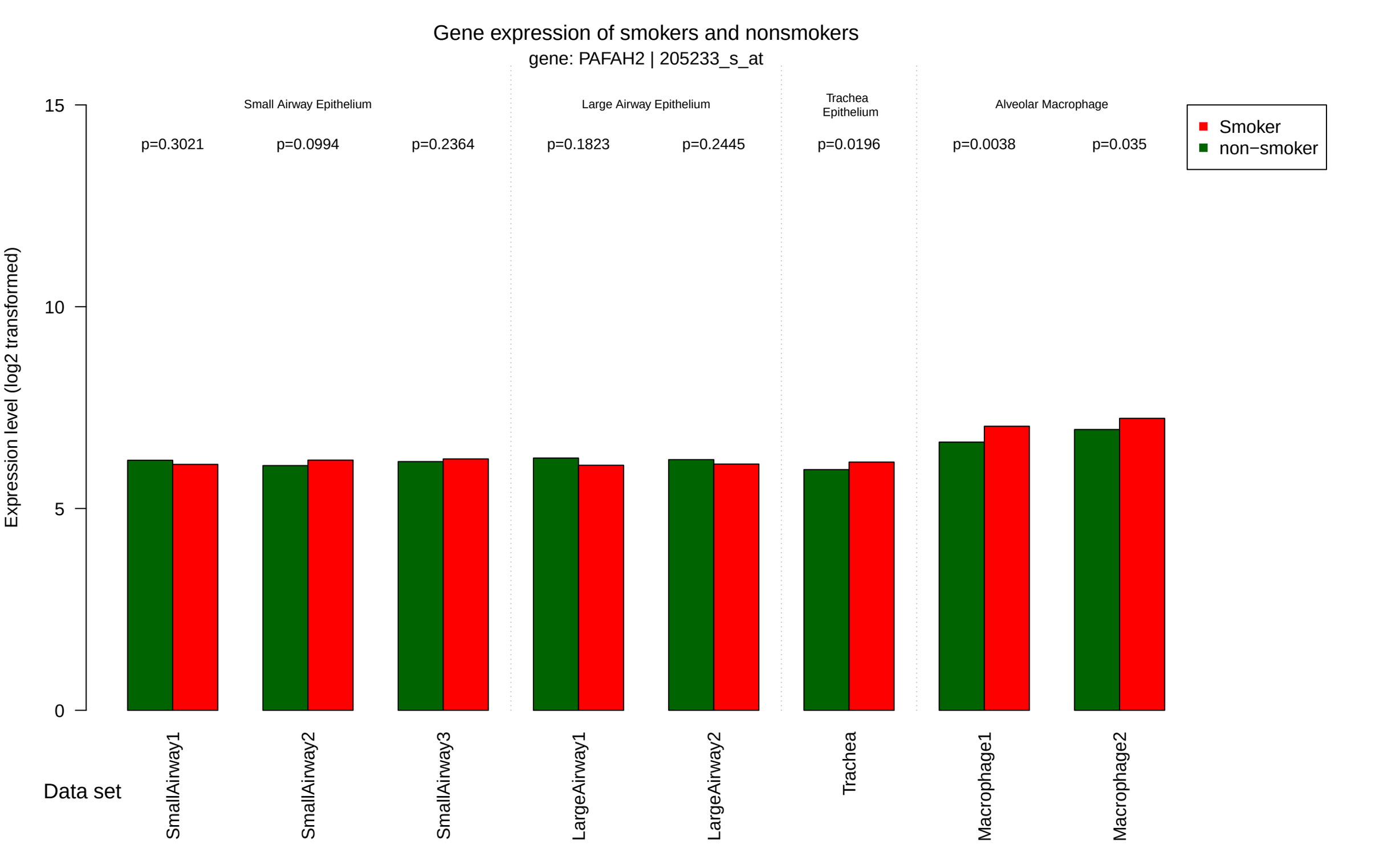
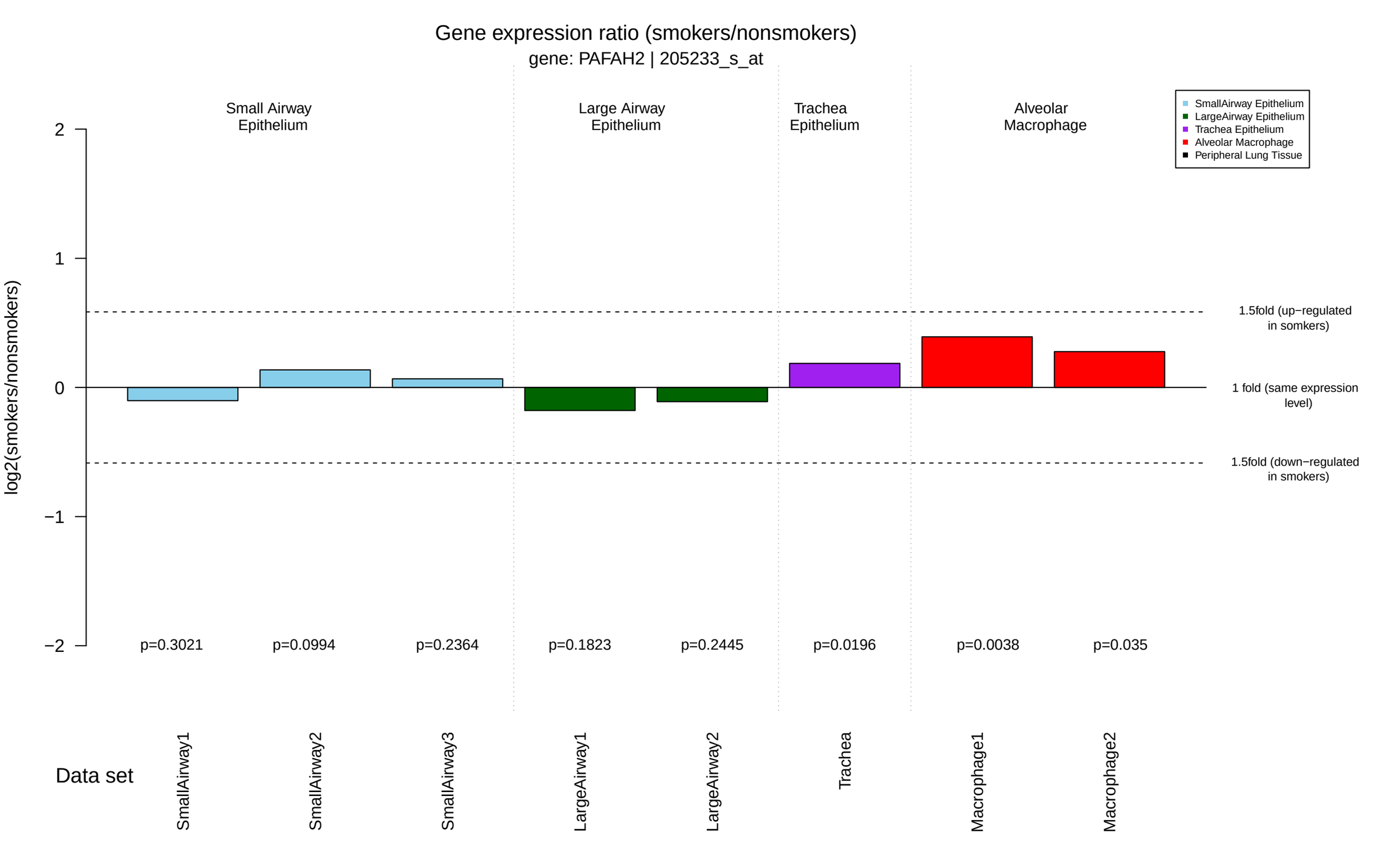
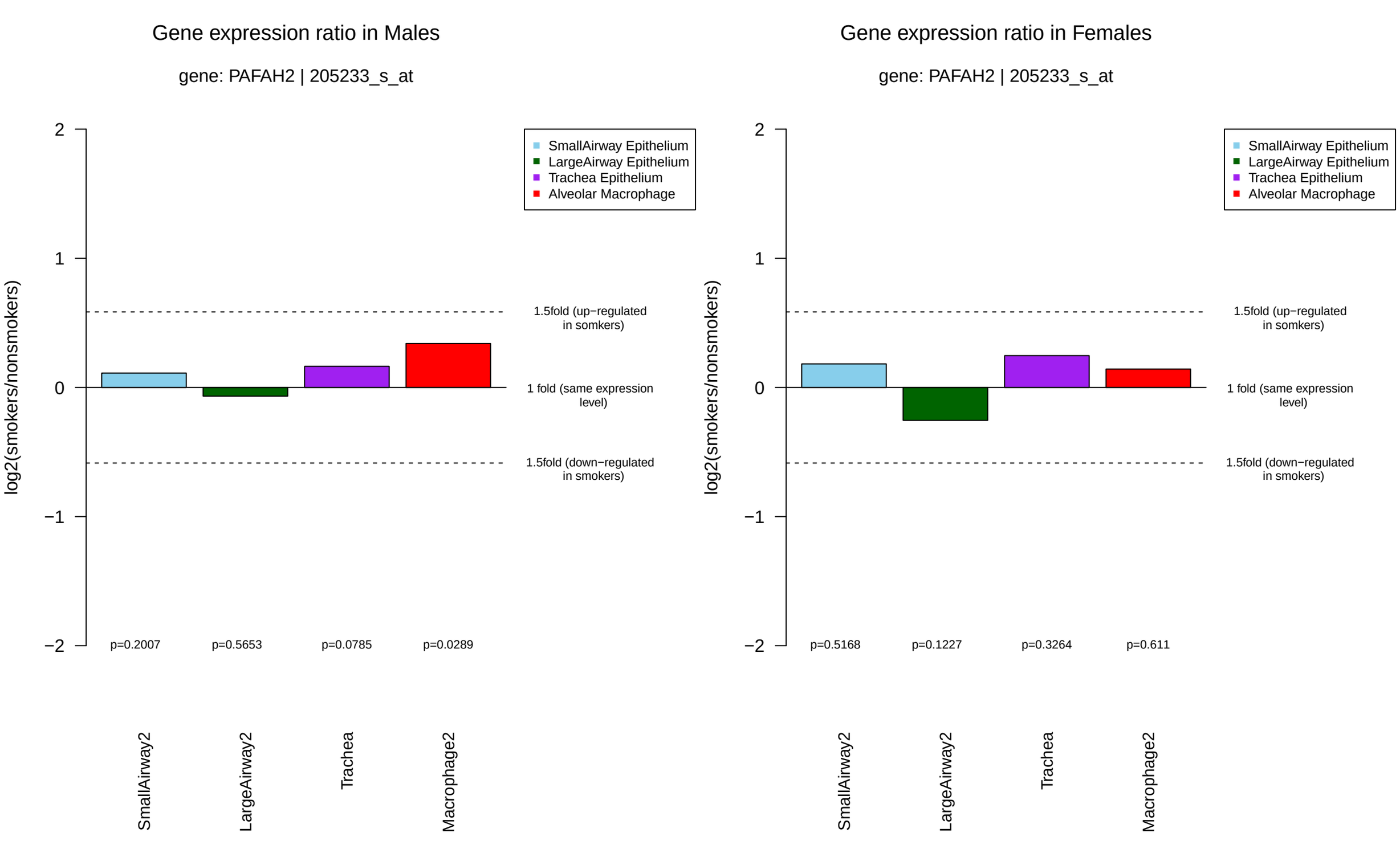
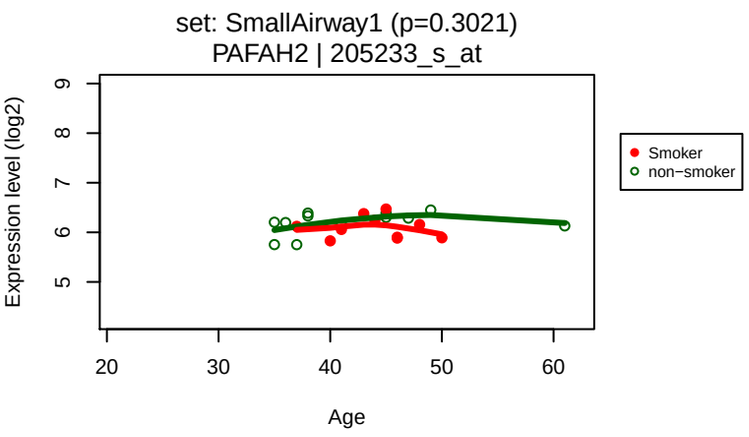
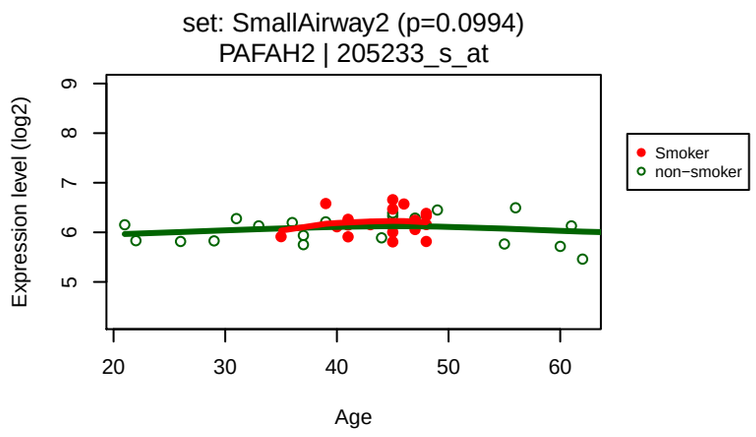
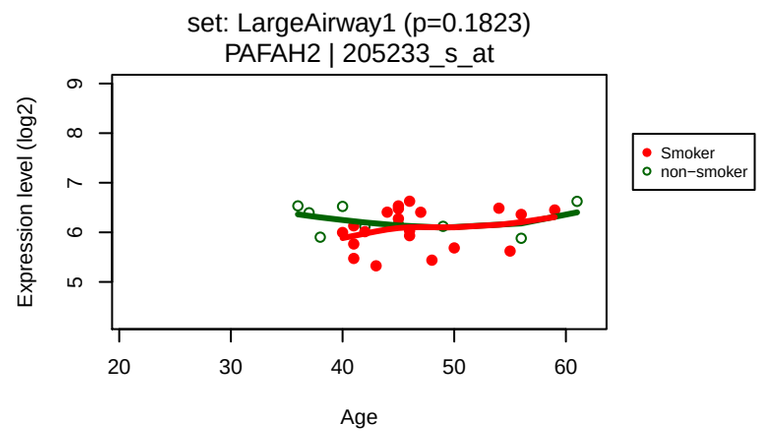

Gene expression of smokers and nonsmokers
gene: PAFAH2 | 205233_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3021
p=0.0994
p=0.2364
p=0.1823
p=0.2445
p=0.0196
p=0.0038
p=0.035
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

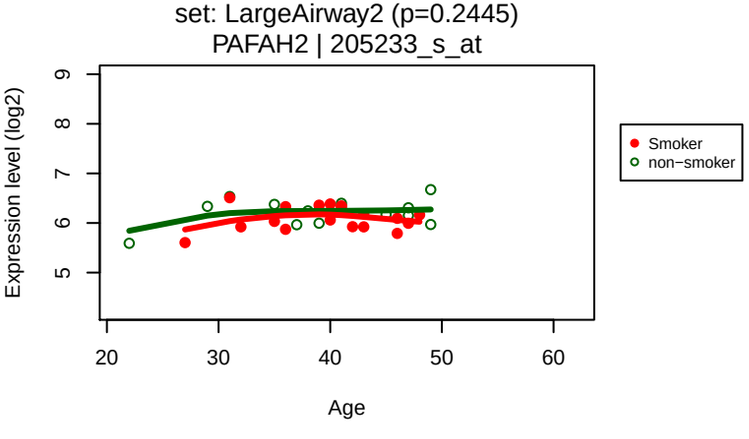

Gene expression ratio (smokers/nonsmokers)
gene: PAFAH2 | 205233_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3021
p=0.0994
p=0.2364
p=0.1823
p=0.2445
p=0.0196
p=0.0038
p=0.035
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

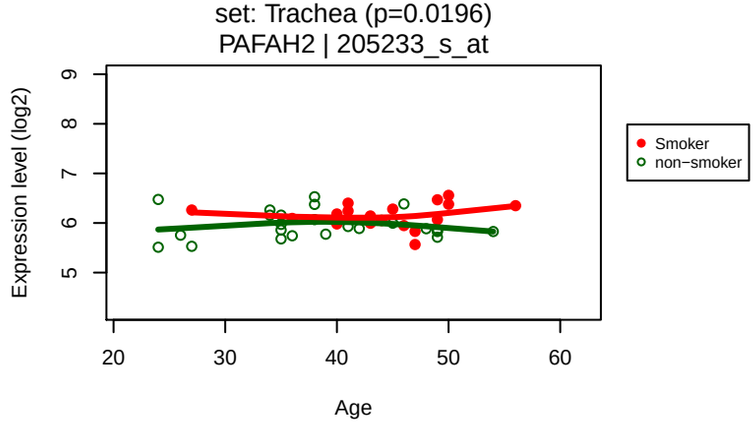

Gene expression ratio in Males
gene: PAFAH2 | 205233_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2007
p=0.5653
p=0.0785
p=0.0289
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAFAH2 | 205233_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5168
p=0.1227
p=0.3264
p=0.611
SmallAirway2
LargeAirway2
Trachea
Macrophage2

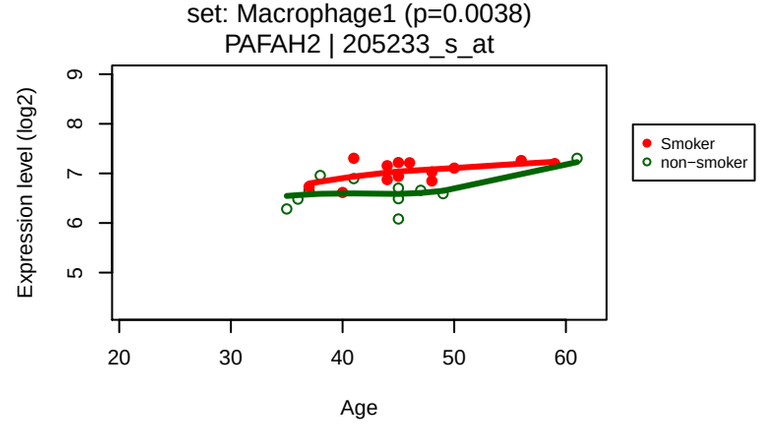

set: SmallAirway1 (p=0.3021)
PAFAH2 | 205233_s_at
Expression level (log2)
Age
Smoker
non−smoker

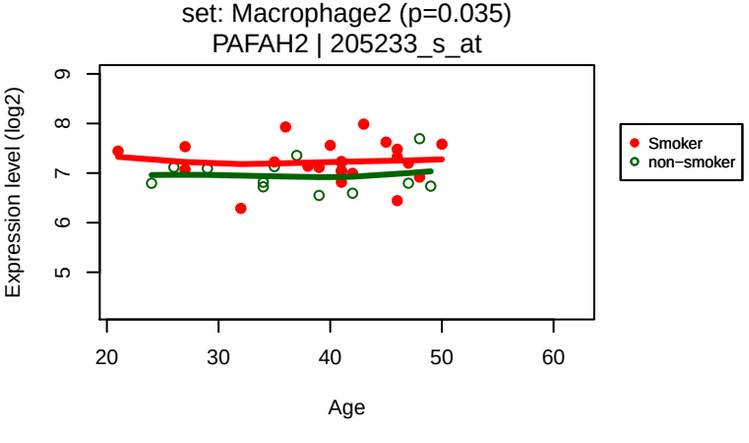

set: SmallAirway2 (p=0.0994)
PAFAH2 | 205233_s_at
Expression level (log2)
Age
Smoker
non−smoker

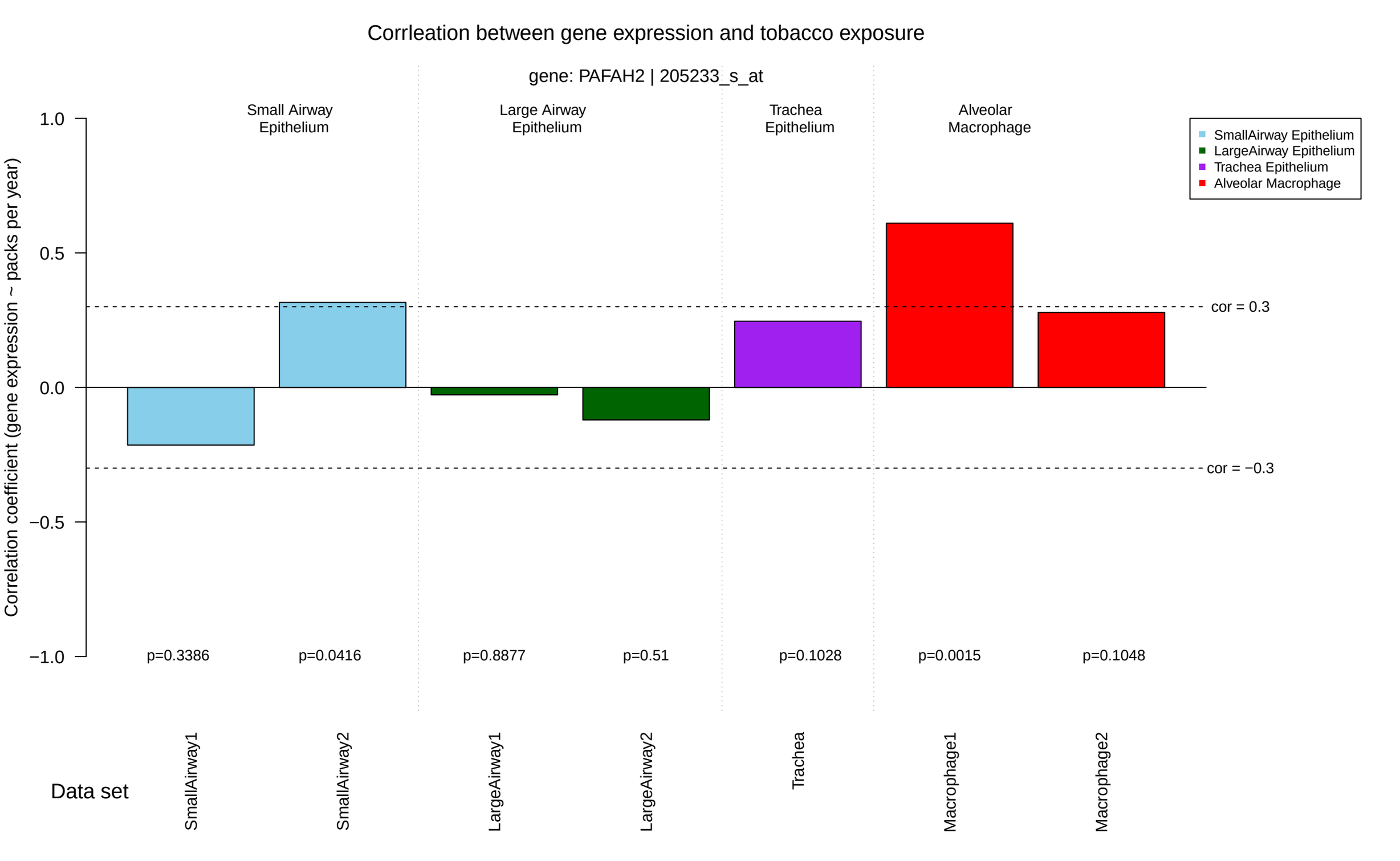

set: LargeAirway1 (p=0.1823)
PAFAH2 | 205233_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2445)
PAFAH2 | 205233_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0196)
PAFAH2 | 205233_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0038)
PAFAH2 | 205233_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.035)
PAFAH2 | 205233_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAFAH2 | 205233_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3386
p=0.0416
p=0.8877
p=0.51
p=0.1028
p=0.0015
p=0.1048
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2